

Credit Risk Prediction Report

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Abstract

This project aims to predict whether a loan applicant is a good or bad credit risk using machine learning models. We compare multiple models and evaluate their performance with a focus on identifying high-risk applicants.

1 Introduction

Credit risk prediction is an essential task in financial decision-making. Lenders must accurately identify applicants who are likely to default in order to minimize financial losses. This project applies machine learning models to classify applicants as either good or bad credit risks.

2 Dataset

We used the UCI German Credit Data, which contains 1,000 observations and 20 features. The dataset classifies applicants into two categories: good or bad credit risks.

The original labels are:

- 1 = Good
- 2 = Bad

We transformed them into:

- 0 = Good credit
- 1 = Bad credit

After applying one-hot encoding to categorical variables, the dataset expanded to 49 features.

3 Methods

We implemented and compared three machine learning models:

- Logistic Regression with `class_weight='balanced'`
- Random Forest with `class_weight='balanced'`
- XGBoost with `scale_pos_weight`

Since bad credit cases represent the minority class, we focused on recall for class 1 as a key evaluation metric.

4 Exploratory Data Analysis

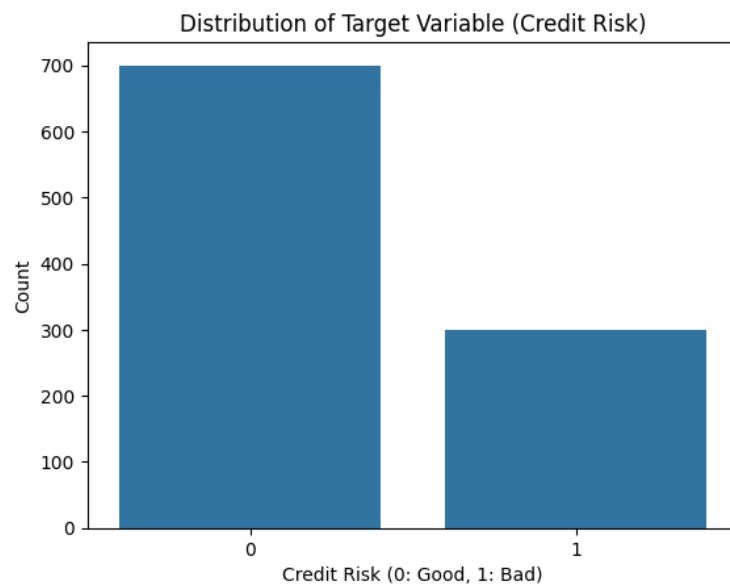


Figure 1: Distribution of Credit Risk

Figure 1 shows the distribution of the target variable. The dataset is imbalanced, with more good credit cases (0) than bad credit cases (1). This imbalance motivates the use of class-weighted models.

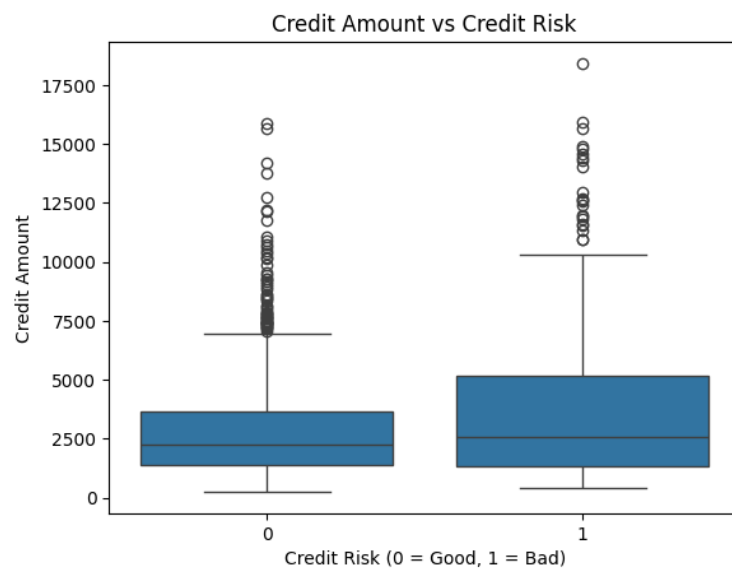


Figure 2: Credit Amount vs Credit Risk

Figure 2 illustrates the relationship between credit amount and credit risk. Higher credit amounts are associated with a higher probability of default.

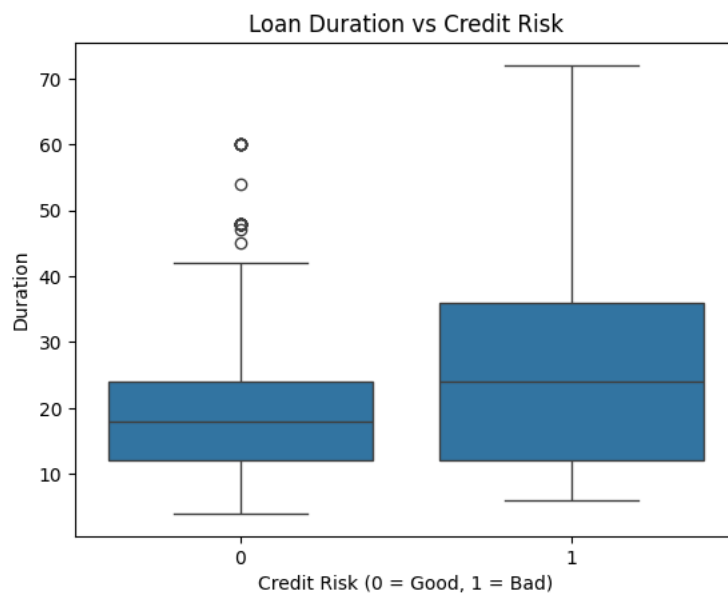


Figure 3: Loan Duration vs Credit Risk

Figure 3 shows that longer loan duration is associated with increased default risk, suggesting that repayment burden plays a key role in credit risk.

Model	Accuracy	ROC-AUC	Recall (Bad Credit)
Logistic Regression	0.750	0.803	0.80
Random Forest	0.775	0.791	0.35
XGBoost	0.715	0.779	0.52

Table 1: Model Performance Comparison

5 Results

Logistic Regression achieved the highest recall for bad credit applicants, indicating it is the most effective model for identifying high-risk individuals.

6 Discussion

In credit risk prediction, false negatives are particularly costly because they represent cases where risky applicants are incorrectly classified as safe. As shown in Figure 1, the dataset is imbalanced, which justifies the use of class-weighted models. Therefore, recall for the bad credit class is more important than overall accuracy.

Although Random Forest achieved the highest accuracy, its recall for bad credit was significantly lower. XGBoost showed moderate performance across all metrics. Logistic Regression provided the best balance between ROC-AUC and recall, making it the most suitable model for this task.

7 Conclusion

The recommended model is Logistic Regression with class weighting. It achieved an ROC-AUC of 0.803 and a recall of 0.80 for bad credit applicants.

Future improvements may include hyperparameter tuning, adjusting classification thresholds, and exploring additional techniques for handling class imbalance.